

Bending, tension, and thermal membrane fluctuations

From the Helfrich energy to a relaxation spectrum

A fluctuating membrane provides two different measurements: its equal-time roughness probes elastic costs, while its relaxation probes dissipation. This note derives that distinction for a nearly flat, passive membrane. The aim is a useful baseline for more elaborate membrane models, not a model of all fluctuations in a living cell.

1 A controlled small-slope model

Consider a fluid membrane with bending modulus κ , Gaussian modulus $\bar{\kappa}$, prescribed tension $\sigma \geq 0$, mean curvature H , Gaussian curvature K , and spontaneous curvature C_0 . The Helfrich energy is [1]

$$F = \int dA \left[\frac{\kappa}{2} (2H - C_0)^2 + \bar{\kappa} K + \sigma \right]. \quad (1)$$

The tension here is the coefficient conjugate to area in the chosen harmonic model. Identifying it with a measured mechanical tension requires care beyond that approximation.

Take a symmetric membrane, $C_0 = 0$, with spatially constant moduli. Write its shape as a height $h(\mathbf{x}, t)$ above a periodic square of side L and projected area $A = L^2$. The expansion requires $|\nabla h| \ll 1$. To quadratic order, $dA \simeq [1 + |\nabla h|^2/2] d^2x$ and $2H \simeq \nabla^2 h$, up to the immaterial normal-orientation sign. For this periodic geometry the Gaussian term does not contribute to the height-dependent quadratic energy. Subtracting the flat-state constant gives

$$F = \frac{1}{2} \int_A d^2x \left[\kappa (\nabla^2 h)^2 + \sigma |\nabla h|^2 \right]. \quad (2)$$

Define $h_{\mathbf{q}} = A^{-1} \int_A h(\mathbf{x}) e^{-i\mathbf{q} \cdot \mathbf{x}} d^2x$, with $\mathbf{q} = 2\pi(n_x, n_y)/L$ and $q = |\mathbf{q}|$. Then

$$h(\mathbf{x}) = \sum_{\mathbf{q}} h_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{x}}, \quad F = \frac{A}{2} \sum_{\mathbf{q} \neq 0} (\kappa q^4 + \sigma q^2) |h_{\mathbf{q}}|^2. \quad (3)$$

Fix the mean height to remove the translation mode $\mathbf{q} = 0$. Because h is real, $h_{-\mathbf{q}} = h_{\mathbf{q}}^*$: opposite wavevectors are not independent complex degrees of freedom.

Equipartition of the independent real sine and cosine amplitudes yields

$$\langle |h_{\mathbf{q}}|^2 \rangle = \frac{k_B T}{A(\kappa q^4 + \sigma q^2)}. \quad (4)$$

Here T is temperature and k_B is Boltzmann's constant. The explicit A is part of the Fourier convention; omitting it changes the inferred modulus. Both terms in the denominator have units of energy per fourth power of length, before multiplication by A .

2 Restoring force and the surrounding fluid

Varying Eq. (2), integrating by parts, and using periodicity gives the normal restoring force per projected area,

$$f = -\frac{\delta F}{\delta h} = -\kappa \nabla^4 h + \sigma \nabla^2 h, \quad f_{\mathbf{q}} = -(\kappa q^4 + \sigma q^2) h_{\mathbf{q}}. \quad (5)$$

Energy specifies a force, but does not specify a time scale. Assume identical incompressible Newtonian fluids of viscosity η on the two sides, unbounded in the normal direction, negligible inertia, no permeation, and no additional membrane or intermonolayer dissipation. Solving the linear Stokes problem for a normal Fourier mode gives the mobility [2]

$$\partial_t h_{\mathbf{q}} = \Lambda(q) f_{\mathbf{q}}, \quad \Lambda(q) = \frac{1}{4\eta q}. \quad (6)$$

The flow penetrates a distance of order q^{-1} into each fluid; viscous traction therefore scales as $\eta q \partial_t h_{\mathbf{q}}$. The factor four is the result for the stated two-sided geometry. It must not be carried unchanged to a supported membrane or a confined fluid.

Adding equilibrium thermal forcing gives

$$\partial_t h_{\mathbf{q}} = -\Gamma(q) h_{\mathbf{q}} + \xi_{\mathbf{q}}, \quad \Gamma(q) = \frac{\kappa q^3 + \sigma q}{4\eta}, \quad (7)$$

$$\langle \xi_{\mathbf{q}}(t) \xi_{\mathbf{q}'}(t') \rangle = \frac{2k_B T \Lambda(q)}{A} \delta_{\mathbf{q}, -\mathbf{q}'} \delta(t - t'). \quad (8)$$

The Gaussian noise has zero mean. Its covariance is fixed by the requirement that the stationary distribution be the Boltzmann distribution of Eq. (2). Indeed the variance balance is $\partial_t \langle |h_{\mathbf{q}}|^2 \rangle = -2\Gamma \langle |h_{\mathbf{q}}|^2 \rangle + 2k_B T \Lambda/A$, reproducing Eq. (4) at stationarity.

Define the two-sided spectrum by $S_h(q, \omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle h_{\mathbf{q}}(t) h_{-\mathbf{q}}(0) \rangle$. For a stationary mode,

$$\langle h_{\mathbf{q}}(t) h_{-\mathbf{q}}(0) \rangle = \frac{k_B T e^{-\Gamma(q)|t|}}{A(\kappa q^4 + \sigma q^2)}, \quad (9)$$

$$S_h(q, \omega) = \frac{2k_B T \Lambda(q)/A}{\Gamma(q)^2 + \omega^2}. \quad (10)$$

The integral of S_h over $d\omega/(2\pi)$ returns the equal-time variance. This is a useful normalization check when comparing temporal and spatial measurements.

3 Crossovers and hidden area

For $\sigma > 0$, bending and tension become comparable at $q = \sqrt{\sigma/\kappa}$. At longer wavelengths the variance scales as q^{-2} and the relaxation rate as q ; at shorter wavelengths they scale as q^{-4} and q^3 . These are model predictions over a range where both the continuum description and the fluid assumptions hold.

The fractional excess area is $\alpha = (A_{\text{actual}} - A)/A \simeq \langle |\nabla h|^2 \rangle / 2$. Replacing the mode sum by a two-dimensional integral gives

$$\alpha \simeq \frac{k_B T}{4\pi} \int_{q_{\min}}^{q_{\max}} \frac{q dq}{\kappa q^2 + \sigma} \quad (11)$$

$$= \frac{k_B T}{8\pi\kappa} \log \frac{\kappa q_{\max}^2 + \sigma}{\kappa q_{\min}^2 + \sigma}. \quad (12)$$

The lower cutoff reflects the system size, roughly $2\pi/L$. The upper cutoff reflects the spatial resolution or breakdown of the continuum membrane model; it is not an arbitrary fitting freedom. Increasing tension suppresses fluctuations and releases hidden area without requiring equivalent molecular stretching. The approximation itself requires $2\alpha \ll 1$.

Numerical check. For illustrative values $T = 300$ K, $\kappa = 20k_B T$, $\sigma = 10^{-7}$ N/m, and $\eta = 10^{-3}$ Pa s, the crossover wavelength $2\pi\sqrt{\kappa/\sigma}$ is $5.72 \mu\text{m}$. Equation (7) gives a relaxation time of 0.189 ms at wavelength $1 \mu\text{m}$, and 48.0 ms at $10 \mu\text{m}$. Wavelength means $2\pi/q$, not $1/q$.

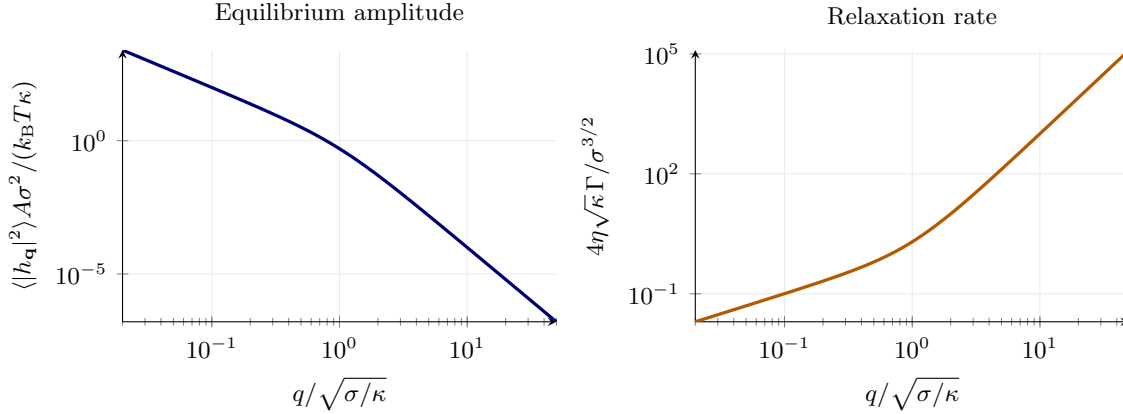


Figure 1: Analytic spectra of the harmonic model, rescaled by the bending–tension crossover. The curves are theoretical, not experimental data. The same crossover has different signatures in equilibrium amplitudes and relaxation rates.

4 What changes in a biological membrane?

A height spectrum alone does not identify the dissipation mechanism. Intermonolayer friction introduces an additional slow mode [2]; confinement changes the fluid mobility; and membrane composition can couple to curvature. Neither those effects nor active forcing can be incorporated reliably by changing the bending modulus alone. For a curved, compositionally varying membrane, the coupled balance laws and entropy production provide a systematic route [3].

A practical fitting sequence is to state the Fourier normalization, establish a spatial range with adequate resolution, infer elastic parameters from static data, and then test the dynamics using those parameters. Failure of the single-exponential prediction can be informative. It is not, by itself, proof of activity: passive hidden variables also generate several time scales. In particular, the passive flat-patch model above is not a complete description of a red blood cell and its cytoskeleton.

References

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